

Simon

The AI Apprentice

ONE-PAGE TECHNICAL REPORT

HACK-NATION × ELEVENLABS

CAPTURE → MAP → TEACH

Challenge Tackled

When experienced staff leave, recordings preserve their clicks—not their judgment. **Simon captures the “why,” limits, and exceptions behind expert decisions**, turns them into a reviewable Work Map, and tutors a new hire on unfamiliar cases. Our prototype serves **accounts-payable experts and new hires**: transfer know-how without asking the expert to write a handbook.

01 · Capture

Screen events + spoken reasoning

02 · Map

Evidence + expert confirmation

03 · Teach

Guidance before the wrong save

Tools / ML Models Used

- **ElevenLabs ElevenAgents + Gemini 2.5 Flash**: interviewer and tutor dialogue. **Scribe v2 Realtime**: transcription. **Turbo v4**: configured speech synthesis.
- **Claude Haiku 4.5**: screen interpretation. **Claude Sonnet 4.5**: structured rule extraction, via **Vercel AI Gateway + AI SDK 7**.
- **Next.js 16, React 19, TypeScript, Tailwind 4**: web interface and APIs. **Zod 4**: schema validation.
- **Supabase PostgreSQL + private Storage**: durable sessions, maps, and media. **Vercel**: deployment.
- **Vitest + Playwright**: regression tests and keyless browser-smoke tooling.

Models listed are repository defaults/configuration; deployment may override them. This is not a live-provider benchmark.

What Worked Well

- **One shared Work Map** connects expert review, evidence replay, and learner guidance.
- **Explicit confirmation** separates a draft from teachable knowledge; the sandbox adds a server-side save guard.
- **A keyless rehearsal path** makes sample maps and teaching explorable without provider credentials; fallbacks are labeled.

What Was Challenging

- **Asking without interrupting**: a deterministic governor gates questions using silence, stillness, and activity signals.
- **Grounding rules in evidence**: verbatim-quote checks, open knowledge slots, and expert corrections reduce invented reasoning.
- **Privacy across async work**: pre-upload masks and consent-version checks prevent stale work from restoring withdrawn evidence.

How We Spent Our Time

Four parallel lanes: voice & timing · screen, privacy & platform · map & tutor · experience & pitch.

1 · Scope

Define the AP scenario, shared contracts, and lane ownership.

2 · Build

Develop voice, screen capture, the Work Map compiler, tutor, and UI in parallel.

3 · Integrate & harden

Connect the workflow; address privacy, stale state, and deployment.

4 · Package

Reserve the final phase for rehearsal, videos, documentation, and submission.

Phase-based summary of the 24-hour build approach; exact hourly allocations were not documented in the repo.

With 24 more hours: test with real experts and new hires, measuring rule fidelity, interruption quality, and independent learning.

Prototype limits: enforced save blocking applies to our sandbox, not arbitrary third-party apps. Live voice quality and learner outcomes need human validation; app deletion does not imply provider deletion.

Demo · tacit-ai-apprentice.vercel.app

Source · github.com/Backpacked333/sorcerer-apprentice-